

Multivariate Models with R

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1 Loading Data

We will use the *Palmer Penguin* data set available directly from the `palmerpenguins` package. You can learn more about the data at allisonhorst.github.io/palmerpenguins.

Today, we will explore the variability in penguin weight.

```
# install.packages("palmerpenguins")
library(palmerpenguins)
library(ggplot2)
library(dplyr)
library(car)

dat <- as.data.frame(penguins) # rename dataset
str(dat) # structure of dataset
```

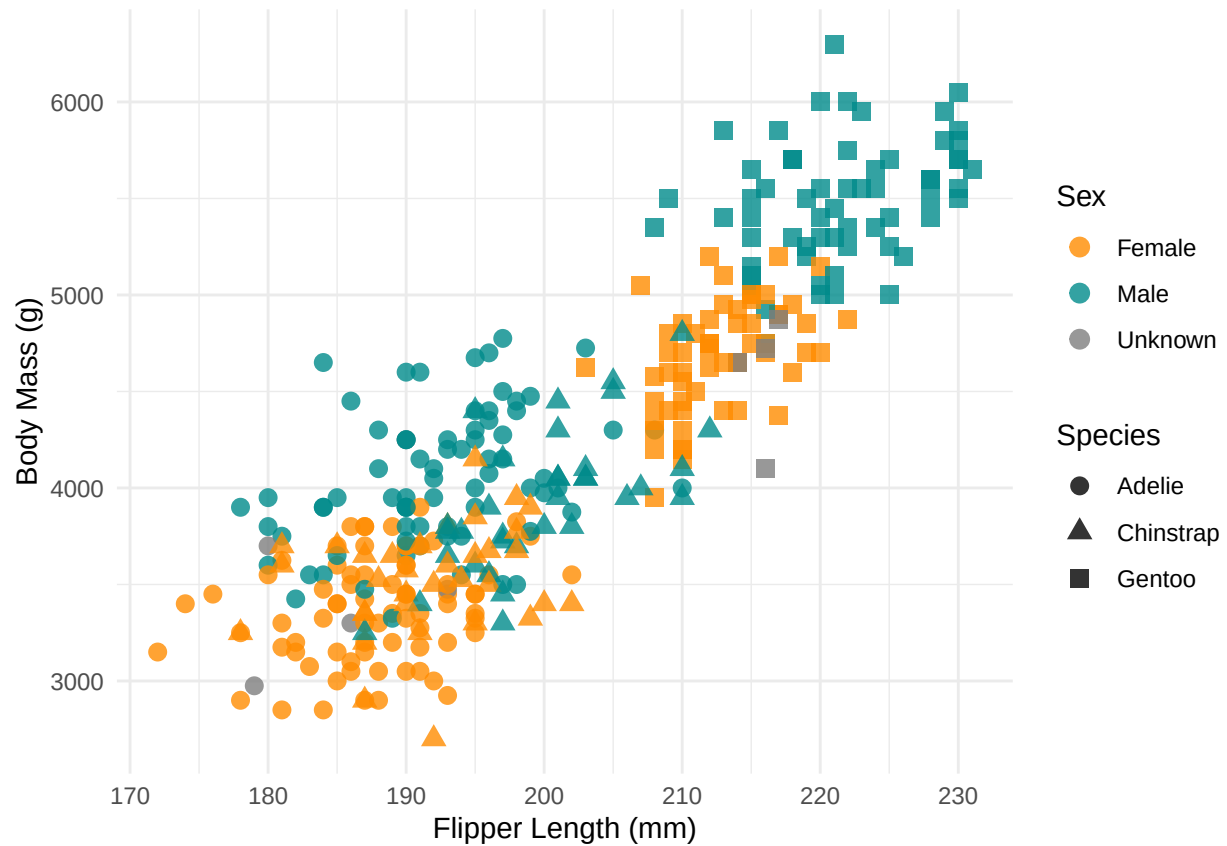
2 Simple Multivariate Models

Several explanatory variables can be included in regression models.

2.1 Visualization

```
ggplot(dat, aes(x = flipper_length_mm, y = body_mass_g, color = sex, shape = species)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                    name = "Sex",
                    labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                    name = "Species") +
  labs(
    x = "Flipper Length (mm)",
    y = "Body Mass (g)"
```

```
) +  
theme_minimal()
```



2.2 Model

```
# Fit model  
mod = lm(body_mass_g ~ flipper_length_mm + sex + species,  
         data = dat)
```

2.3 Assumptions

```
#
```

2.4 Inference

```
# Return outputs (including p-value)  
summary(mod)  
  
##  
## Call:  
## lm(formula = body_mass_g ~ flipper_length_mm + sex + species,  
##     data = dat)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max
```

```
## -721.8 -195.7 -5.9 198.6 873.7
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -365.817    532.050  -0.688   0.4922
## flipper_length_mm  20.025      2.846   7.037 1.15e-11 ***
## sexmale        530.381     37.810  14.027 < 2e-16 ***
## speciesChinstrap -87.634     46.347  -1.891   0.0595 .
## speciesGentoo   836.260     85.185   9.817 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 295.6 on 328 degrees of freedom
## (11 observations deleted due to missingness)
## Multiple R-squared:  0.8669, Adjusted R-squared:  0.8653
## F-statistic: 534 on 4 and 328 DF, p-value: < 2.2e-16
```

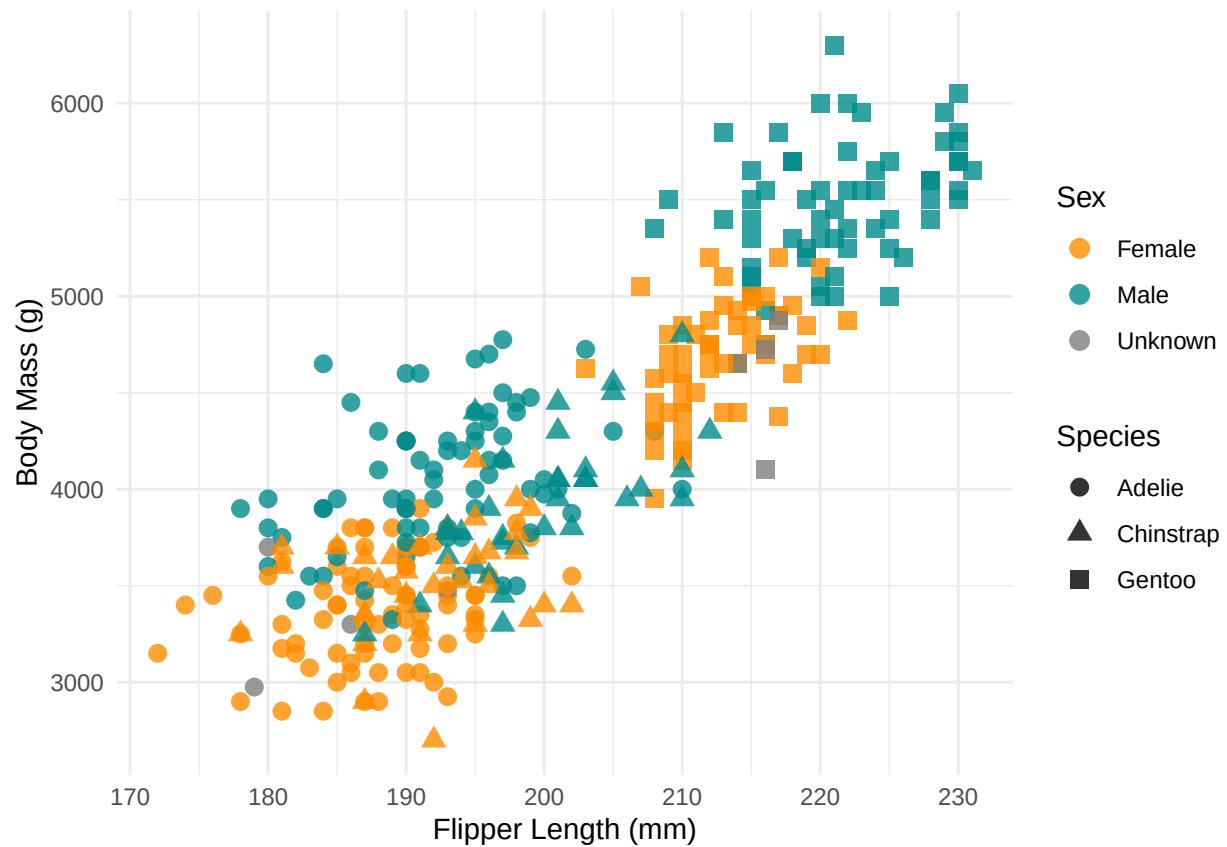
Interpretation...

3 More Complex Multivariate Models

Several explanatory variables can be included in regression models.

3.1 Visualization

```
ggplot(dat, aes(x = flipper_length_mm, y = body_mass_g, color = sex, shape = species)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                    name = "Sex",
                    labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                    name = "Species") +
  labs(
    x = "Flipper Length (mm)",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



Let's focus on gentoo penguins

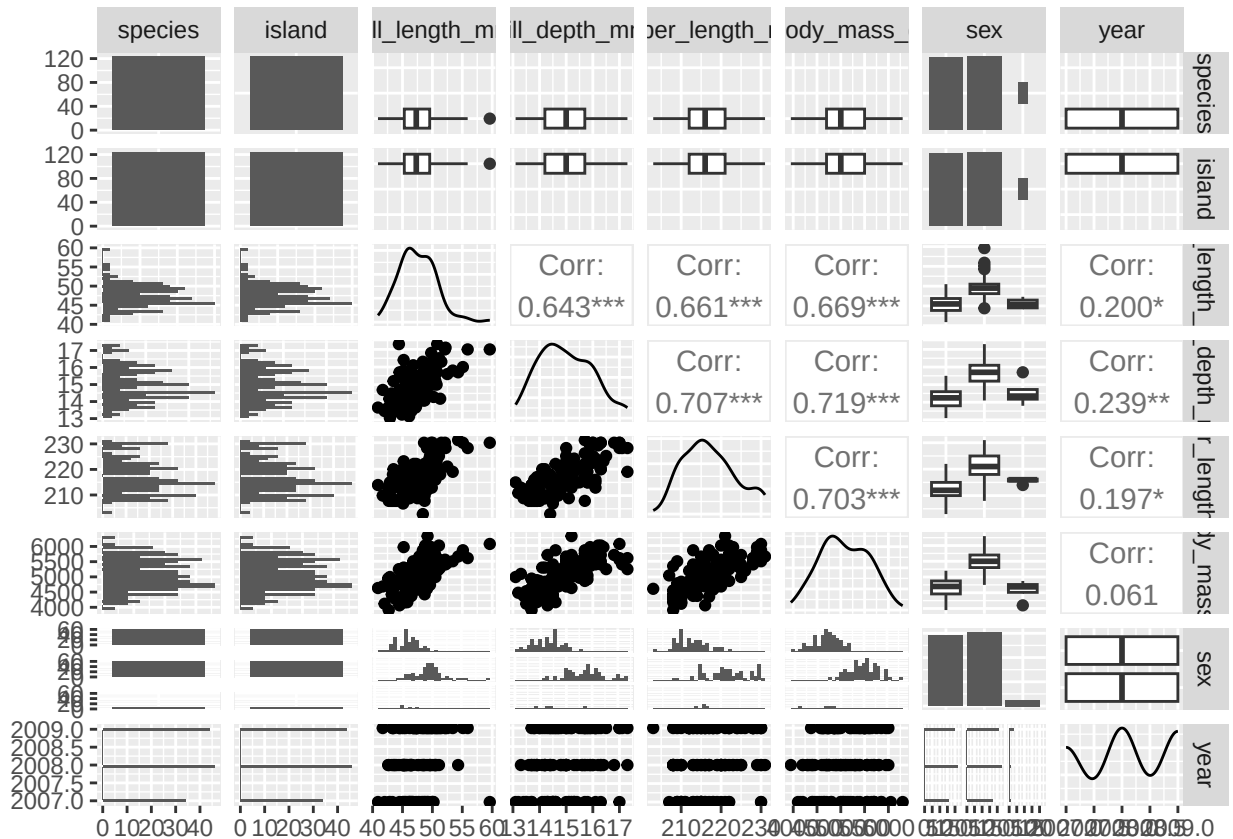
3.2 Subset

```
dat_gep = subset(dat, species == "Gentoo")
```

3.3 Assumptions part 1

3.3.1 Linearity

```
library(GGally)
ggpairs(dat_gep)
```



3.4 Model

```
# Fit model
mod = lm(body_mass_g ~ flipper_length_mm + sex + bill_length_mm + bill_depth_mm + year,
         data = dat_gep)
```

3.5 Assumptions part 2

Collinearity

```
# VIF
vif(mod)

## flipper_length_mm      sex      bill_length_mm      bill_depth_mm
##          2.489679      2.963962          2.139011          3.163876
##              year
##          1.201103
```

3.6 Model Selection

```
mod = lm(body_mass_g ~ flipper_length_mm,
         data = dat_gep)

summary(mod)
```

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm, data = dat_gep)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -911.18 -235.76  -51.93   170.75 1015.71
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -6787.281    1092.552   -6.212 7.65e-09 ***
## flipper_length_mm    54.623      5.028   10.863 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 360.2 on 121 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.4937, Adjusted R-squared:  0.4896
## F-statistic: 118 on 1 and 121 DF, p-value: < 2.2e-16

mod = lm(body_mass_g ~ flipper_length_mm + sex,
          data = dat_gep)

summary(mod)

##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm + sex, data = dat_gep)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -619.13 -182.37    5.87   171.12   827.88
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -318.691    1112.019   -0.287  0.775
## flipper_length_mm    23.499      5.225    4.497 1.64e-05 ***
## sexmale         597.501      68.552    8.716 2.38e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 276.3 on 116 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.7015, Adjusted R-squared:  0.6963
## F-statistic: 136.3 on 2 and 116 DF, p-value: < 2.2e-16

mod = lm(body_mass_g ~ flipper_length_mm + sex + bill_length_mm,
          data = dat_gep)

summary(mod)

##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm + sex + bill_length_mm,
##     data = dat_gep)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -639.20 -173.00   -4.07  171.61  832.30
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -366.241    1090.204  -0.336  0.73753
## flipper_length_mm  17.881      5.634   3.174  0.00193 **
## sexmale         540.502     71.291   7.582 9.54e-12 ***
## bill_length_mm    27.271     11.394   2.394  0.01831 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 270.9 on 115 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.7156, Adjusted R-squared:  0.7082
## F-statistic: 96.47 on 3 and 115 DF,  p-value: < 2.2e-16

mod = lm(body_mass_g ~ flipper_length_mm + sex + bill_length_mm + bill_depth_mm,
         data = dat_gep)

summary(mod)

##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm + sex + bill_length_mm +
##      bill_depth_mm, data = dat_gep)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -611.89 -166.99   -8.53  146.53  859.44
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -496.327    1091.308  -0.455  0.6501
## flipper_length_mm  15.390      5.929   2.596  0.0107 *
## sexmale         491.235     80.375   6.112 1.41e-08 ***
## bill_length_mm    24.041     11.622   2.069  0.0409 *
## bill_depth_mm     56.688     43.197   1.312  0.1920
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 270 on 114 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.7199, Adjusted R-squared:  0.71
## F-statistic: 73.24 on 4 and 114 DF,  p-value: < 2.2e-16

mod = lm(body_mass_g ~ flipper_length_mm + sex + bill_length_mm + year,
         data = dat_gep)

summary(mod)

##
## Call:
```

```
## lm(formula = body_mass_g ~ flipper_length_mm + sex + bill_length_mm +
##     year, data = dat_gep)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -637.08 -177.91    1.83  165.89  811.18
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   40554.32   66870.05   0.606  0.54541
## flipper_length_mm    18.54     5.75   3.224  0.00165 **
## sexmale         529.84    73.58   7.201 6.86e-11 ***
## bill_length_mm     28.50    11.60   2.457  0.01552 *
## year          -20.48    33.45  -0.612  0.54174
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 271.6 on 114 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.7166, Adjusted R-squared:  0.7066
## F-statistic: 72.05 on 4 and 114 DF, p-value: < 2.2e-16
```

Final model

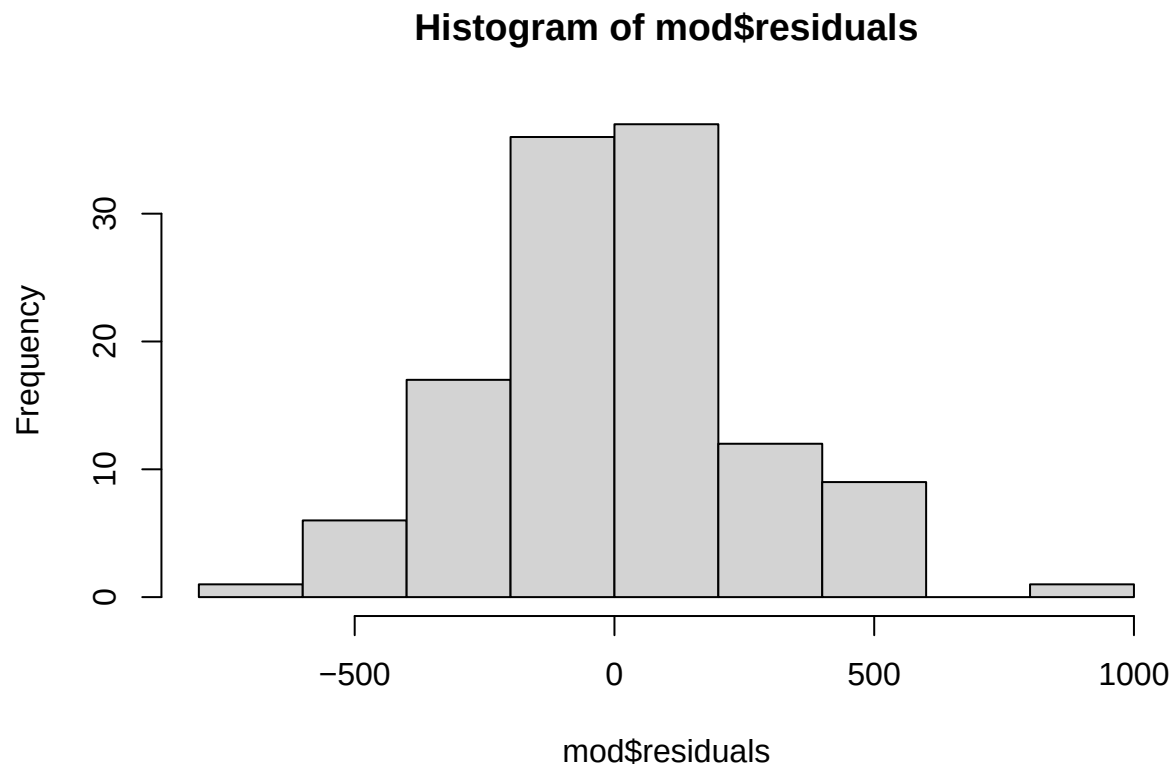
```
mod = lm(body_mass_g ~ flipper_length_mm + sex + bill_length_mm,
         data = dat_gep)
```

```
summary(mod)
```

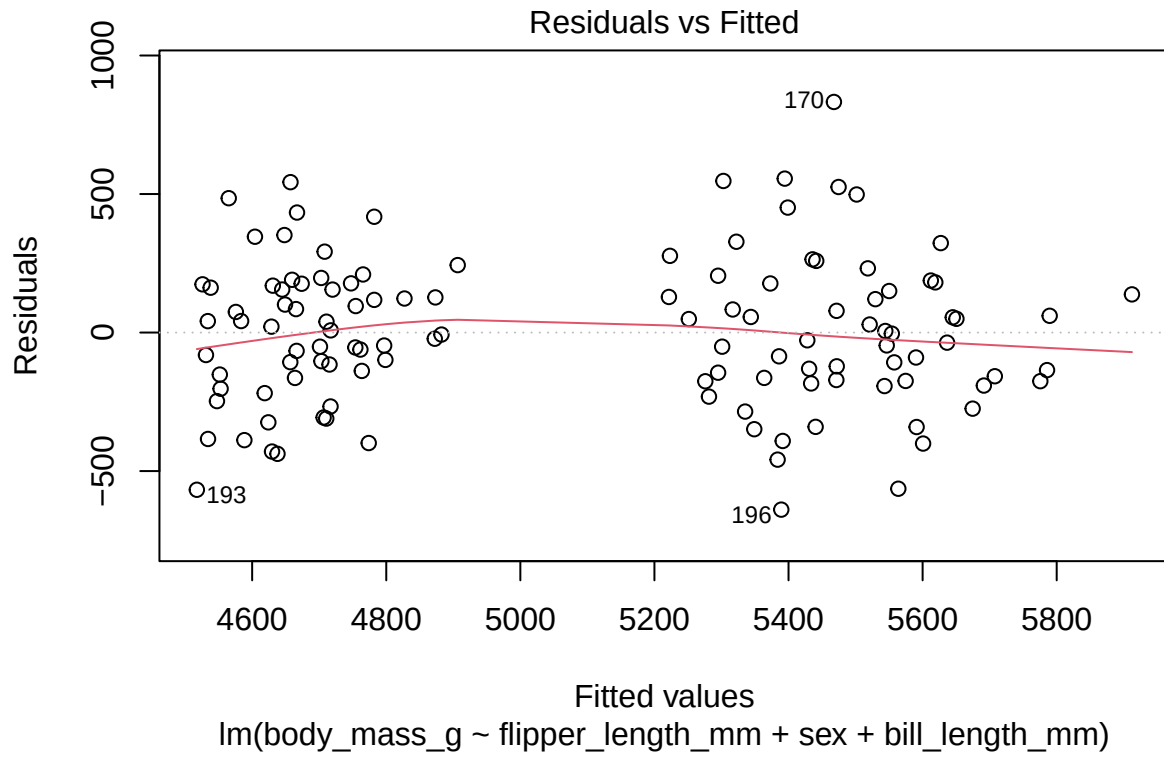
```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm + sex + bill_length_mm,
##     data = dat_gep)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -639.20 -173.00   -4.07  171.61  832.30
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -366.241   1090.204  -0.336  0.73753
## flipper_length_mm    17.881     5.634   3.174  0.00193 **
## sexmale         540.502    71.291   7.582 9.54e-12 ***
## bill_length_mm     27.271    11.394   2.394  0.01831 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 270.9 on 115 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.7156, Adjusted R-squared:  0.7082
## F-statistic: 96.47 on 3 and 115 DF, p-value: < 2.2e-16
```

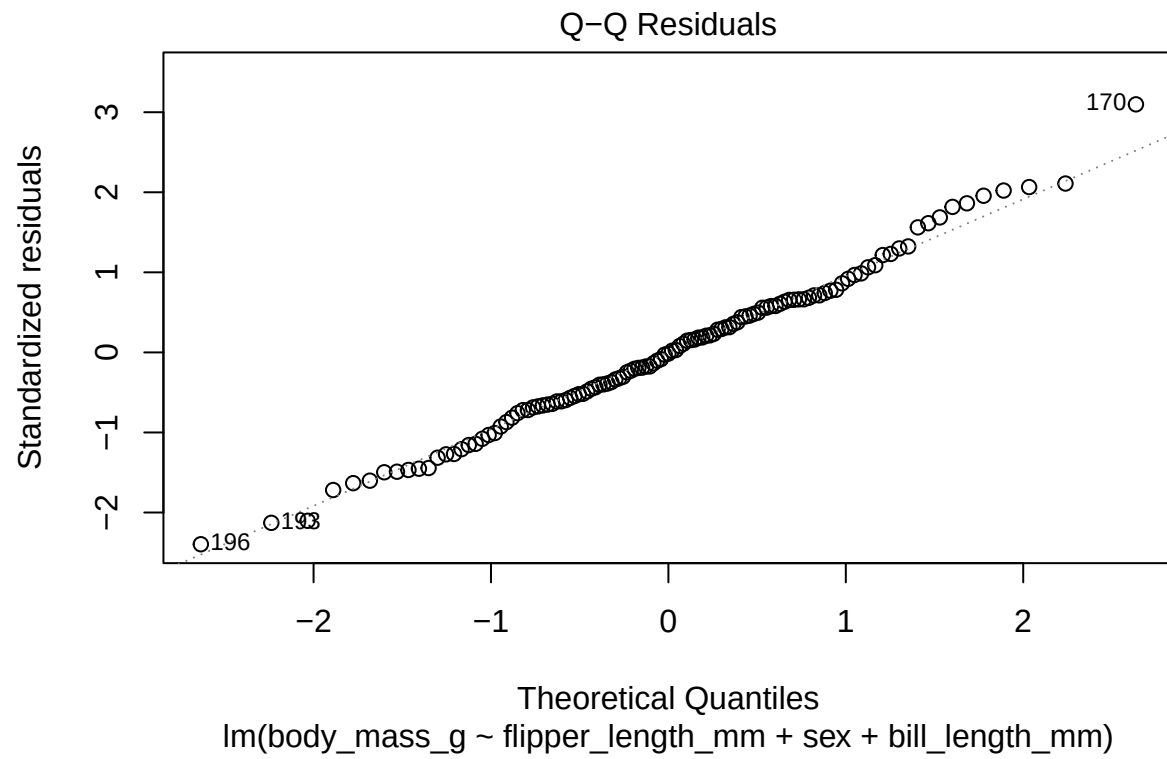

3.7 Assumptions part 3

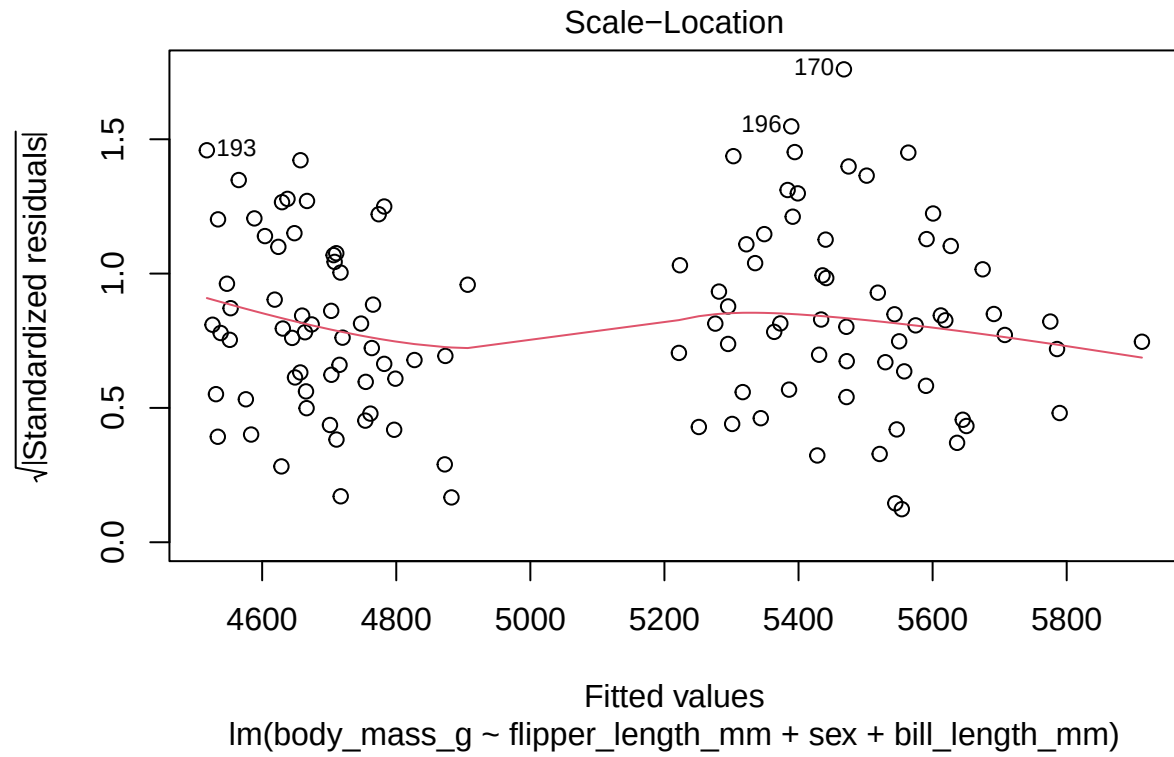
```
hist(mod$residuals)
```

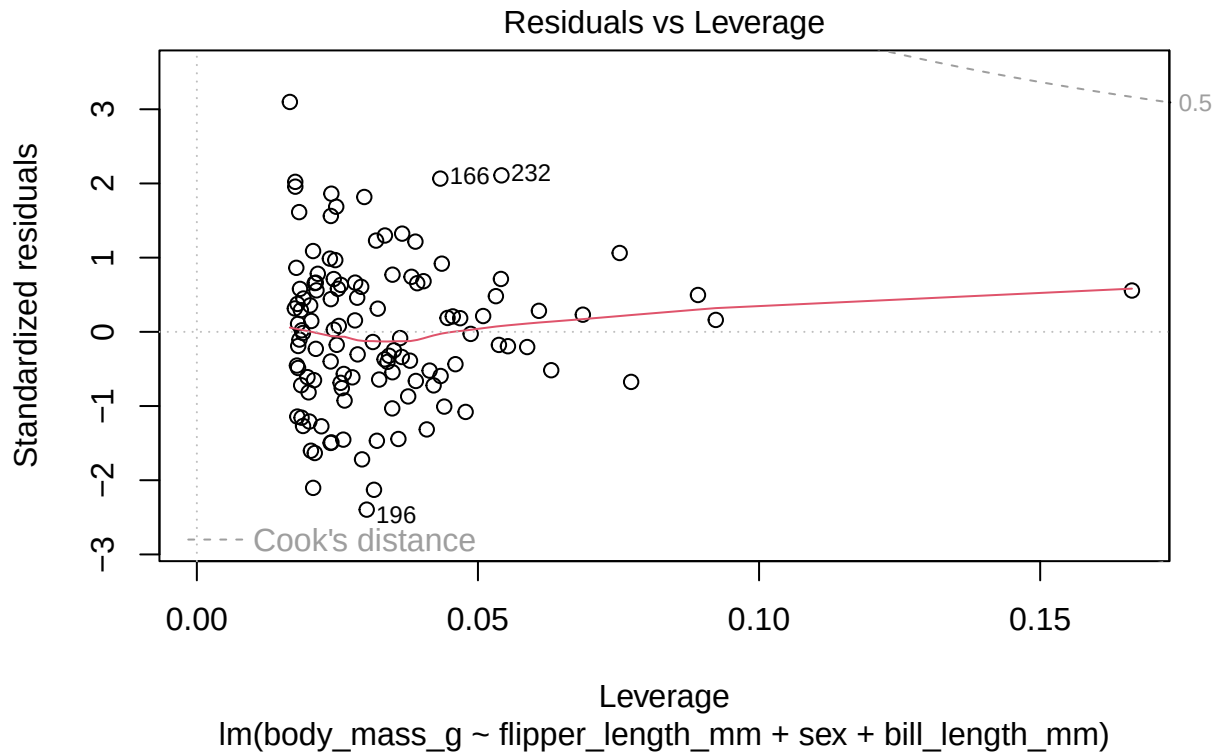


```
plot(mod)
```









3.8 Inference

```
# Return outputs (including p-value)
summary(mod)
```

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm + sex + bill_length_mm,
##     data = dat_gep)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -639.20 -173.00  -4.07   171.61   832.30
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -366.241    1090.204  -0.336  0.73753
## flipper_length_mm    17.881      5.634   3.174  0.00193 **
## sexmale          540.502     71.291   7.582 9.54e-12 ***
## bill_length_mm     27.271     11.394   2.394  0.01831 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 270.9 on 115 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.7156, Adjusted R-squared:  0.7082
```

```
## F-statistic: 96.47 on 3 and 115 DF,  p-value: < 2.2e-16
```

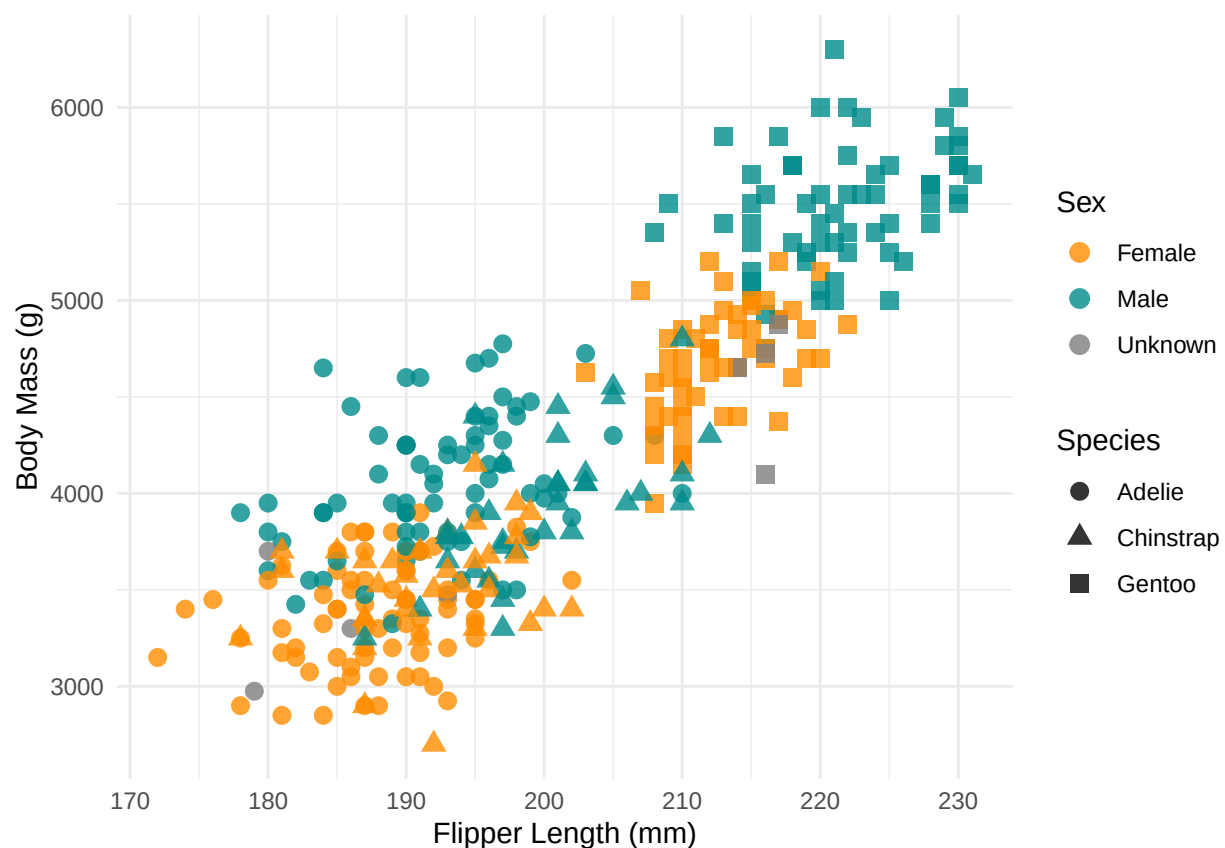
4 Interactions

Here we are looking at the association between weight and 2 categorical variables: sex and species.

Several explanatory variables can be included in regression models.

4.1 Visualization

```
ggplot(dat, aes(x = flipper_length_mm, y = body_mass_g, color = sex, shape = species)) +  
  geom_point(size = 3, alpha = 0.8) +  
  scale_color_manual(values = c("darkorange", "cyan4"),  
                    name = "Sex",  
                    labels = c("Female", "Male", "Unknown")) +  
  scale_shape_manual(values = c(16, 17, 15),  
                    name = "Species") +  
  labs(  
    x = "Flipper Length (mm)",  
    y = "Body Mass (g)"  
  ) +  
  theme_minimal()
```



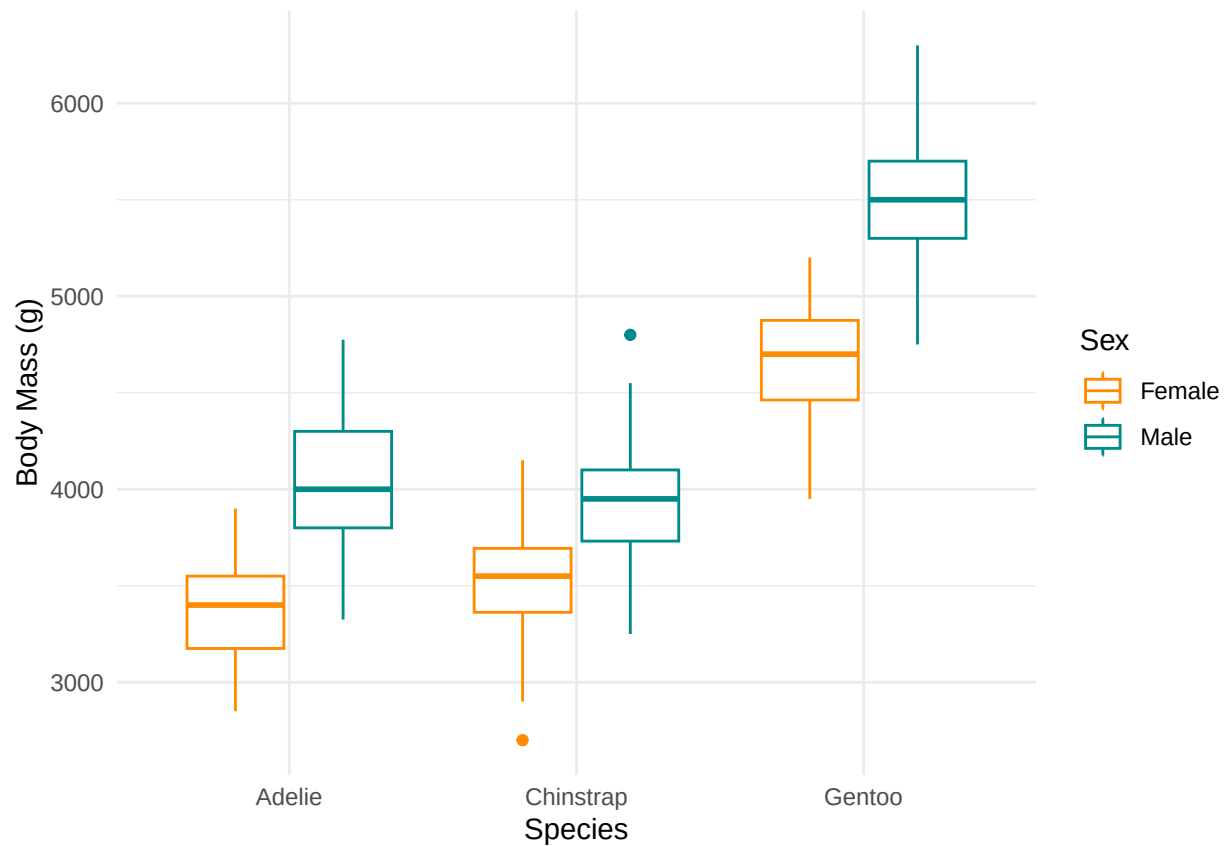
Let's look at the effect of sex across species

```
ggplot(subset(dat, !is.na(sex)), aes(x = species, y = body_mass_g, color = sex)) +  
  geom_boxplot() +
```

```

scale_color_manual(values = c("darkorange", "cyan4"),
                   name = "Sex",
                   labels = c("Female", "Male", "Unknown")) +
labs(
  x = "Species",
  y = "Body Mass (g)"
) +
theme_minimal()

```



4.2 Model Selection

```

mod = lm(body_mass_g ~ species,
          data = dat)

summary(mod)

##
## Call:
## lm(formula = body_mass_g ~ species, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1126.02  -333.09   -33.09   316.91  1223.98
##
## Coefficients:

```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3700.66     37.62   98.37  <2e-16 ***
## speciesChinstrap    32.43     67.51    0.48   0.631
## speciesGentoo    1375.35     56.15   24.50  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 462.3 on 339 degrees of freedom
## (2 observations deleted due to missingness)
## Multiple R-squared:  0.6697, Adjusted R-squared:  0.6677
## F-statistic: 343.6 on 2 and 339 DF, p-value: < 2.2e-16
```

```
mod = lm(body_mass_g ~ species + sex,
          data = dat)
```

```
summary(mod)
```

```
##
## Call:
## lm(formula = body_mass_g ~ species + sex, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -816.87 -217.80  -16.87   227.61   882.20
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3372.39     31.43  107.308  <2e-16 ***
## speciesChinstrap    26.92     46.48    0.579   0.563
## speciesGentoo    1377.86     39.10   35.236  <2e-16 ***
## sexmale          667.56     34.70   19.236  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 316.6 on 329 degrees of freedom
## (11 observations deleted due to missingness)
## Multiple R-squared:  0.8468, Adjusted R-squared:  0.8454
## F-statistic: 606.1 on 3 and 329 DF, p-value: < 2.2e-16
```

```
mod = lm(body_mass_g ~ species + sex + sex:species,
          data = dat)
```

```
summary(mod)
```

```
##
## Call:
## lm(formula = body_mass_g ~ species + sex + sex:species, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -827.21 -213.97   11.03   206.51   861.03
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3368.84     36.21   93.030  < 2e-16 ***
```



```
## speciesChinstrap      158.37      64.24    2.465  0.01420 *
## speciesGentoo         1310.91     54.42   24.088 < 2e-16 ***
## sexmale                674.66     51.21   13.174 < 2e-16 ***
## speciesChinstrap:sexmale -262.89     90.85   -2.894  0.00406 **
## speciesGentoo:sexmale   130.44     76.44    1.706  0.08886 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 309.4 on 327 degrees of freedom
## (11 observations deleted due to missingness)
## Multiple R-squared:  0.8546, Adjusted R-squared:  0.8524
## F-statistic: 384.3 on 5 and 327 DF,  p-value: < 2.2e-16

Final model
mod = lm(body_mass_g ~ species + sex + sex:species,
          data = dat)

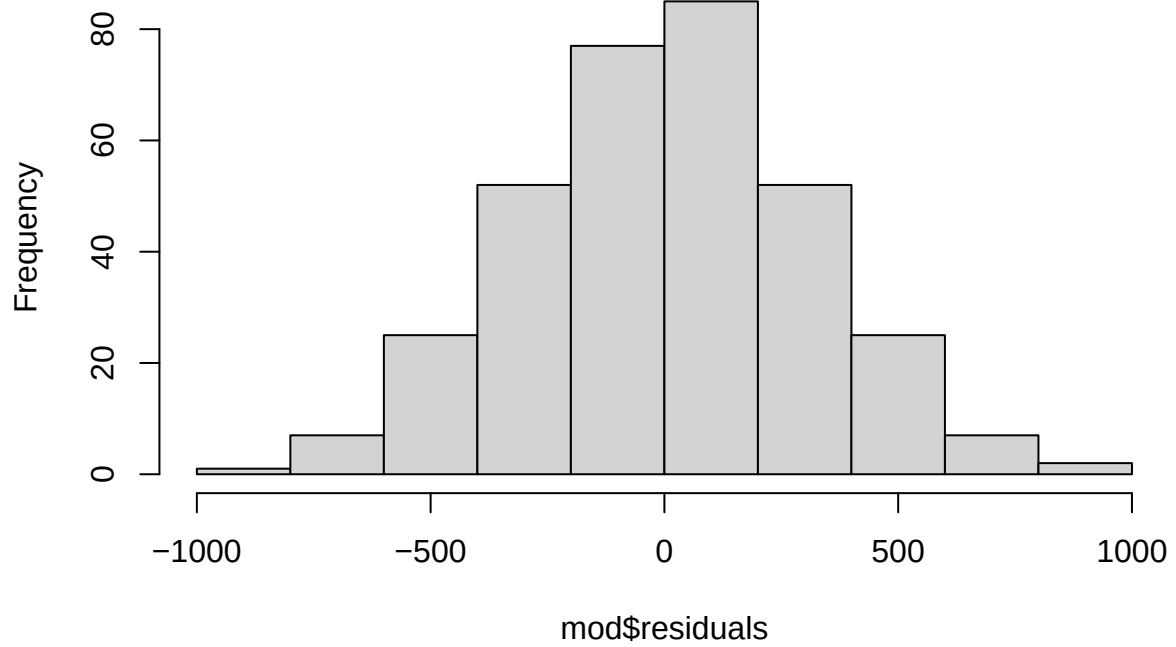
summary(mod)

##
## Call:
## lm(formula = body_mass_g ~ species + sex + sex:species, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -827.21 -213.97   11.03  206.51  861.03
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3368.84      36.21  93.030 < 2e-16 ***
## speciesChinstrap    158.37      64.24   2.465  0.01420 *
## speciesGentoo      1310.91     54.42  24.088 < 2e-16 ***
## sexmale            674.66     51.21  13.174 < 2e-16 ***
## speciesChinstrap:sexmale -262.89     90.85   -2.894  0.00406 **
## speciesGentoo:sexmale   130.44     76.44    1.706  0.08886 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 309.4 on 327 degrees of freedom
## (11 observations deleted due to missingness)
## Multiple R-squared:  0.8546, Adjusted R-squared:  0.8524
## F-statistic: 384.3 on 5 and 327 DF,  p-value: < 2.2e-16
```

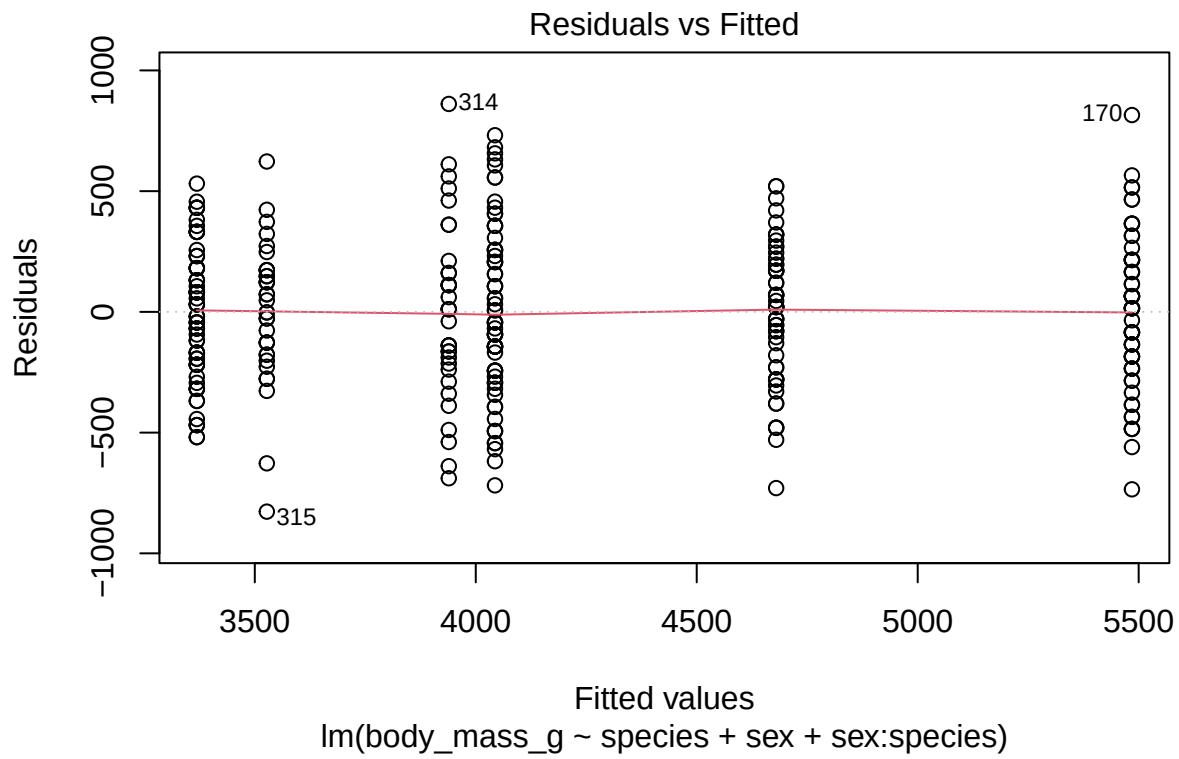
4.3 Assumptions

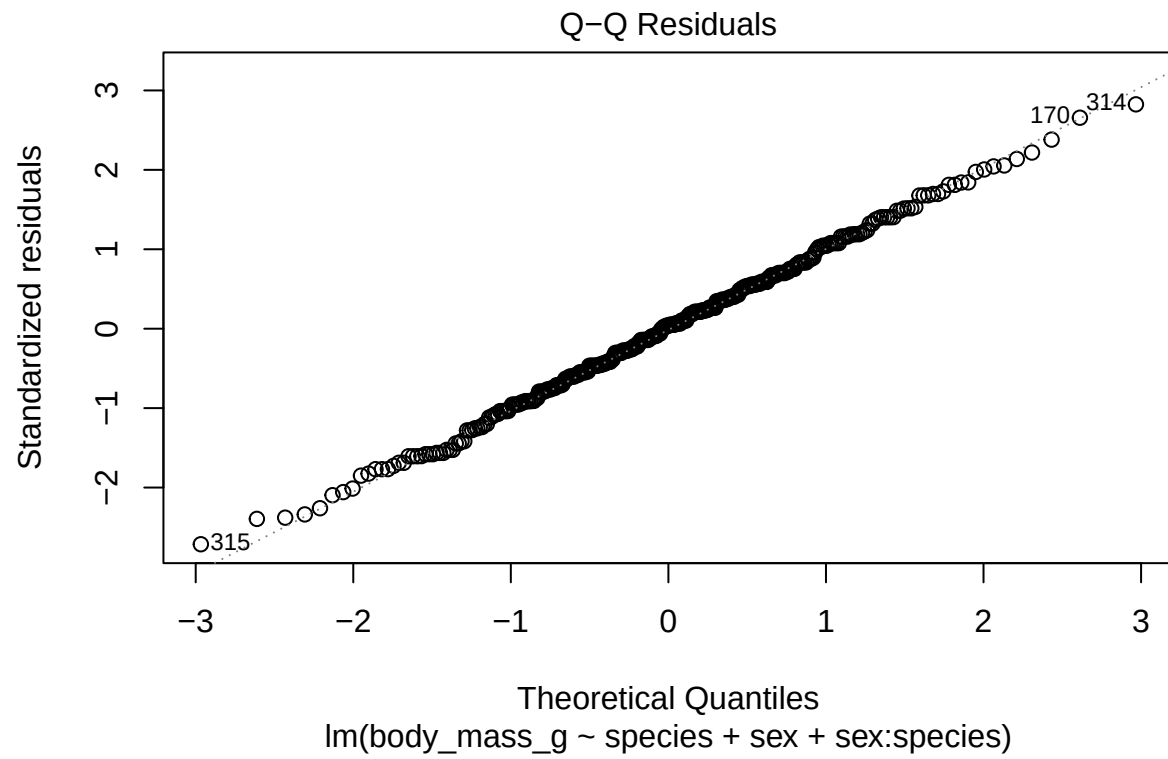
```
hist(mod$residuals)
```

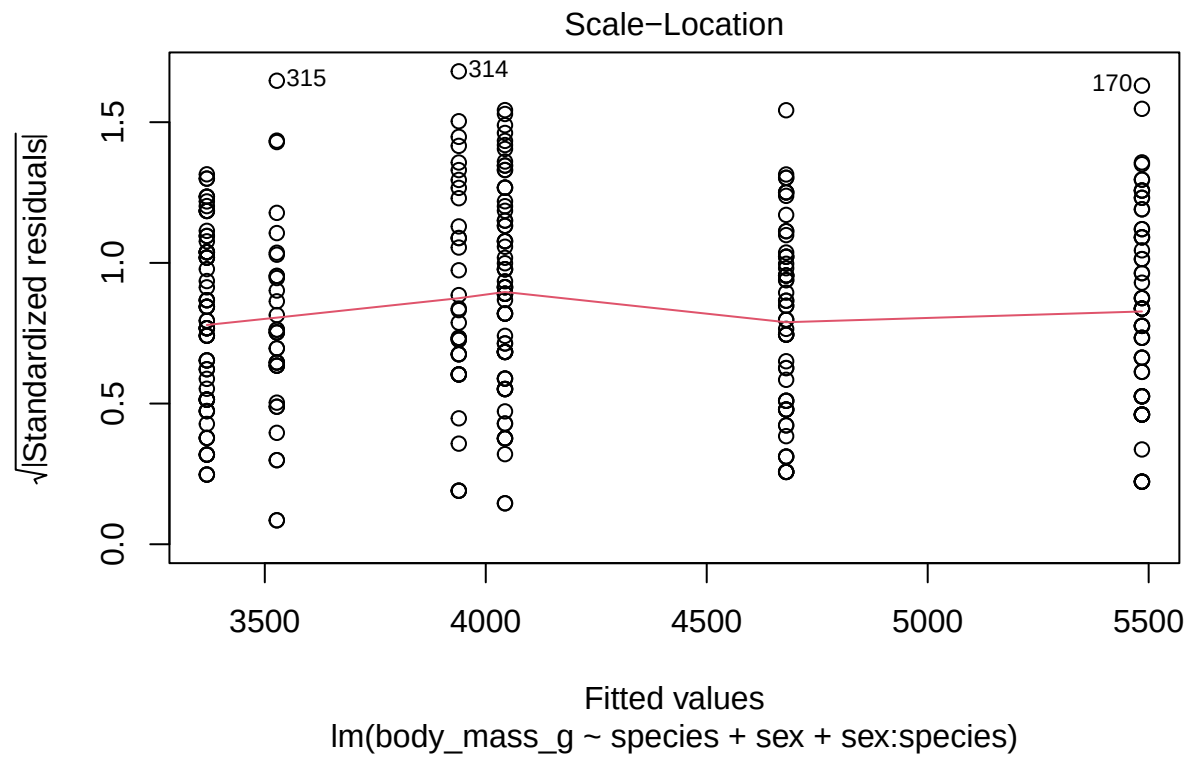
Histogram of mod\$residuals

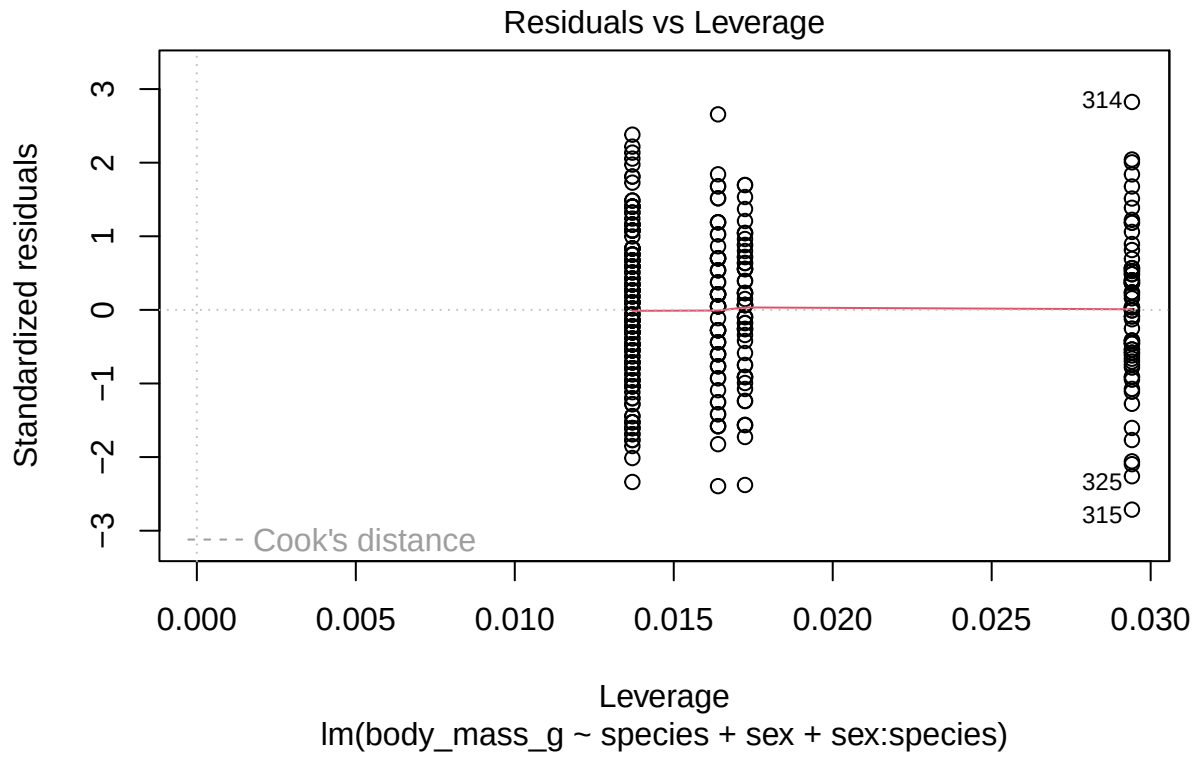


```
plot(mod)
```









4.4 Inference

```
# Return outputs (including p-value)
summary(mod)
```

```
##
## Call:
## lm(formula = body_mass_g ~ species + sex + sex:species, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -827.21 -213.97   11.03  206.51  861.03
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3368.84     36.21  93.030 < 2e-16 ***
## speciesChinstrap    158.37     64.24   2.465  0.01420 *
## speciesGentoo     1310.91     54.42  24.088 < 2e-16 ***
## sexmale           674.66     51.21  13.174 < 2e-16 ***
## speciesChinstrap:sexmale -262.89     90.85  -2.894  0.00406 **
## speciesGentoo:sexmale   130.44     76.44   1.706  0.08886 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 309.4 on 327 degrees of freedom
## (11 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.8546, Adjusted R-squared:  0.8524  
## F-statistic: 384.3 on 5 and 327 DF,  p-value: < 2.2e-16
```